RIS-Assisted ISAC Systems for Robust Secure Transmission With Imperfect Sense Estimation

Chengjun Jiang, *Graduate Student Member, IEEE*, Chensi Zhang, *Member, IEEE*, Chongwen Huang, *Member, IEEE*, Jianhua Ge, Dusit Niyato, *Fellow, IEEE*, and Chau Yuen, *Fellow, IEEE*

***Abstract*— In this paper, we investigate reconfigurable intelligent surface (RIS)-assisted integrated sensing and communication (ISAC) systems for robust physical layer security (PLS) schemes. Traditionally, eavesdroppers (Eves) typically avoid interacting with the base station, making it challenging to obtain their relevant information, which limits the implementation of PLS. Fortunately, the sensing information obtained by ISAC can contribute to the design of PLS. Therefore, leveraging imperfect sensing estimation and employing dedicated radar signals as artificial noise, we formulate an RIS-assisted joint active and passive beamforming design problem to maximize the sum secrecy rate while satisfying the user's quality of service constraints, the transmission power constraints, and the sensing signal strength requirements. To make the problem tractable, we first derive the bound for Eve's channel state information uncertainty region based on security approximations. Subsequently, we employ the *$\mathcal{S}$-procedure* and the symbolic-deterministic methods to transform the infinite number of inequalities. We then utilize the first-order Taylor expansion, the second-order cone methods, and the successive convex approximation to address the nonconvexity problem, leading to an efficient suboptimal solution obtained by an iterative algorithm. Finally, the simulation results demonstrate the significant potential of the sensing function in enhancing security and the effectiveness of the proposed robust scheme in flexibly balancing communication and sensing quality.**

***Index Terms*— Integrated sensing and communication, reconfigurable intelligent surface, physical layer security.**

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Chengjun Jiang, Chensi Zhang, and Jianhua Ge are with the State Key Laboratory of Integrated Service Networks, Xidian University, Xi'an 710071, China (e-mail: cjjiang@stu.xidian.edu.cn; cszhang@xidian.edu.cn; jhge@xidian.edu.cn).

Chongwen Huang is with the College of Information Science and Electronic Engineering, Zhejiang University, Hangzhou 310027, China, and also with Zhejiang Provincial Key Laboratory of Multi-Modal Communication Networks and Intelligent Information Processing, Hangzhou 310027, China (e-mail: chongwenhuang@zju.edu.cn).

Dusit Niyato is with the College of Computing and Data Science, Nanyang Technological University, Singapore 639798 (e-mail: dniyato@ntu.edu.sg).

Chau Yuen is with the School of Electrical and Electronics Engineering, Nanyang Technological University, Singapore 639798 (e-mail: chau.yuen@ntu.edu.sg).

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I. Introduction

As WIRELESS technology continues to evolve, the demand for both reliable communication and high-precision sensing services in next-generation wireless systems is becoming increasingly desirable. Traditionally, sensing and communication systems are designed separately, occupying different frequency bands. However, this approach leads to doubled resource overheads, including spectrum and hardware resources. As both radio sensing and communication systems transition to higher frequency bands, larger antenna arrays, and miniaturization, their hardware architecture, channel characteristics, and signal processing methods have become increasingly similar [1]. Consequently, integrated sensing and communications (ISAC) offers an exciting opportunity to share resources, thereby enhancing spectral efficiency and reducing hardware costs [2]. In light of this, ISAC, as a cornerstone technology in next-generation wireless communications, has garnered extensive attention [3], [4], [5], [6], [7], [8], [9].

To extend the application scenarios and enhance the system performance of ISAC, it is integrated with advanced wireless technologies. For example, multiple-input multiple-output (MIMO) is widely adopted in ISAC, which leverages the degrees of freedom (DoF) of spatial architecture to significantly improve waveform diversity for radar sensing while achieving spatial multiplexing and beamforming gain for communication [10]. Therein, designing transmit waveforms and beamforming for ISAC systems is a crucial issue [11]. In [5], Liu et al. proposed an MIMO-based beamforming design for joint radar sensing and multiuser communication. Obviously, the impact of DoF on the performance of ISAC systems is critical, and thus the need for antenna array size is more urgent for a dual-function radar communication (DFRC) base station (BS) compared to the single-function BS. However, large-scale antenna arrays demand advanced signal processing capabilities to manage and refine data from numerous antennas, potentially resulting in increased processing delays and heightened system complexity. Besides that, performance degradation also remains inevitable under harsh propagation conditions [12].

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Therefore, reconfigurable intelligent surfaces (RIS) offers a potential solution [13], [14], [15], [16], [17], [18], [19], [20].

RIS consists of numerous passive components that manipulate the propagation path of electromagnetic waves without requiring extra energy. This passive nature offers several advantages, including reduced energy consumption, the elimination of complex signal processing, increased deployment flexibility, etc [21]. Moreover, RIS can enhance the DoF of beam design while providing additional line-of-sight (LoS) links to counteract complex electromagnetic environments. In view of this, the application of RIS in ISAC systems has been widely investigated. Furthermore, RIS deployment options vary based on the specific demands for communication and radar sensing [22], [23], [24], [25], [26], [27]. For example, the RIS can be deployed near the user solely to assist in communications [22], [23]; it can be positioned where the direct link is obstructed to establish the reflective link [24], [25], or it can utilize both direct and reflective links to support communications and radar sensing [26], [27].

Due to the broadcast nature of the wireless medium, wireless communication systems are inherently vulnerable to potential security breaches [28], [29], [30]. To this end, Wyner proposed as early as 1975 that if the channel quality of the primary is better than that of the eavesdropping, perfect secure transmission in the presence of the eavesdropper (Eve) can be achieved [31]. In light of this, alongside cryptographic techniques, physical layer security (PLS) as an important complementary technology can enhance the security of wireless systems. However, the eavesdropping properties of Eves make it challenging to obtain information about them, which limits the implementation of PLS. The introduction of ISAC offers a unique opportunity to mitigate this challenge by utilizing its sensing capability to actively estimate Eve information, such as distance, angle, and channel conditions [32]. Specifically, ISAC enables the dual use of radar sensing and communication signals to illuminate potential Eves, thereby obtaining critical sensing information from the echoes. This sensing-based CSI estimation can then be leveraged to implement more effective PLS schemes, even when the Eves are passive.

A. Related Works

The implementation of the ISAC for PLS has garnered significant attention, and related studies can be categorized into two types based on whether they are used in conjunction with RIS.

1) ISAC for PLS [33], [34], [35], [36], [37]: Assuming the targets were Eves and its imperfect channel state information (CSI) is accessible, Liu et al. proposed a robust scheme in ISAC systems that guaranteed both secure communication and sensing services [33]. Given that Eve's CSI was available, Chu et al. explored the use of strong radar signals to ensure PLS in multi-user multiple-input single-output and MIMO radar coexistence systems [34]. In the aforementioned works, the radar and secure communication operated independently rather than cooperatively. In fact, a prerequisite for the implementation of many PLS schemes is the need to obtain specific information about Eve, e.g., the CSI and the direction, etc., which often renders PLS impractical. Leveraging sensing capabilities in ISAC, Jia et al. proposed a robust optimization model for PLS in [35]. In [36], Xu et al. investigated a robust resource allocation design for sense-assisted secure communication in the ISAC system, where the recognized targets were considered potential Eves, and the CSI of Eves was obtained based on the sensing information. Likewise, Su et al. investigated the ISAC systems for sense-assisted PLS in [37], i.e., utilizing the sensing function of ISAC by estimating information about potential Eves.

2) RIS-Assisted ISAC for PLS [38], [39], [40]: RIS is unquestionably beneficial to the ISAC system, offering additional DoF and broader coverage capabilities. Supposing the target was a potential Eve and its CSI was accessible, Jiang et al. deployed an RIS to assist the communicating user, thereby enhancing system security [38]. For the two cases where the target location is perfectly and imperfectly known, Hua et al. designed two corresponding strategies to enhance the PLS of the RIS-aided ISAC system in [39], in which RIS not only assists multiple users in downlink communication but also creates LoS links for target sensing. Considering that a target's entire message can be perfectly captured in [40], Jiang et al. utilize the additional DoF provided by the RIS to enhance secure communication and implement target detection within the non-orthogonal multiple access-assisted ISAC system.

B. Motivation and Contributions

The RIS-assisted ISAC system provides additional design DoFs, which significantly aids in the design of PLS when the information regarding the Eve is assumed to be known. However, the main limitation affecting the implementation of PLS is the requirement to obtain accurate information about potential Eves. Thus, using sensing capabilities to collect potential Eve's information is a feasible approach to assist secure communication [36], [37]. It is worth noting that in existing studies on secure RIS-assisted ISAC transmission, the information about Eve is often assumed to be perfectly known and idealized. In fact, although an RIS and a BS with multi-dimensional sensing capabilities can obtain more accurate target information compared to systems without an RIS, estimation errors are still unavoidable. To this end, we utilize imperfect sensing estimates obtained through ISAC to develop a robust beamforming design that ensures reliable communication and sensing tracking functions. This approach offers a significant practical advantage over traditional methods that typically assume perfect CSI or lack mechanisms to acquire it. This is the focus of the research in this paper, and the contributions of our work are summarized as follows:

In our work, we utilize imperfect sensing estimates obtained through ISAC to develop a robust beamforming design that balances both sensing accuracy and secure communication quality, which provides a significant advancement over traditional methods that typically assume perfect CSI or lack mechanisms to acquire it.

- We investigate the robust PLS schemes in RIS-assisted ISAC systems. Specifically, we consider that only the target sensing parameters estimated by the sensing function

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are available. Due to hardware limitations, unavoidable estimation errors, etc., the system can only rely on distance and angle values that contain inaccuracies. In addition, we employ the dedicated radar signals as artificial noise for joint sensing and PLS configuration. Then, we propose the sum secrecy rate maximization problem to satisfy the user's quality of service (QoS) constraints, transmission power constraints, sensing signal strength requirements.

- To address this problem, we first derive an operational boundary based on security approximations to capture the angle and distance uncertainty of Eve. Subsequently, we employ the *$\mathcal{S}$-procedure* and the symbolic-deterministic methods to handle the infinite number of inequalities and further rewrite them as the finite number of linear matrix inequalities (LMI). To address the nonconvexity of the problem, we utilize the first-order Taylor expansions, second-order cone (SOC), and successive convex approximation (SCA). Building on this, we propose an iterative algorithm based on alternating optimization (AO) framework.
- Our simulation results validate the effectiveness of the proposed robust scheme in achieving a flexible trade-off between communication quality and target sensing quality. Additionally, significant potential is also demonstrated for the sensing function in enhancing security within ISAC systems.

C. Organization

This paper is organized as follows. We analyze the model of RIS-assisted ISAC systems in Section II. The preformance metrics and problem formation are considered in Section III. In Section IV, the bounds for Eve's CSI uncertainty region and problem solution are given. Simulation results are given in Section V. Finally, conclusions are provided in Section VI.

Notations: $(\cdot)^{\dagger}$, $(\cdot)^{T}$, and $(\cdot)^{*}$ represent the conjugate transpose, transpose, and conjugate, respectively. $\mathrm{diag}\,(\cdot)$ is diagonal matrix with main diagonal $(\cdot)$. $|\cdot|$ and $\|\cdot\|_2$ represent the absolute value of scalar and 2-norm of complex vector, respectively. $A \triangleq B$ means that A is defined as B. $\mathbb{C}^{a\times b}$ denotes $a \times b$ complex matrices. $x \sim \mathcal{CN}\,(u, v)$ represents that the complex Gaussian variable with mean u and variance v. $\mathfrak{Re}\,(s)$ and $\mathfrak{Im}\,(s)$ are the real and imaginary parts of the complex scalar s, respectively. $\otimes$ denotes the Kronecker product operator, and $[x]^{+} = \max\,(x, 0)$.

II. System Model

We consider a model of RIS-assisted ISAC as shown in Fig. 1, where the DFRC BS serves K users while detecting an Eve location with the assistance of RIS. The BS has N transmission antennas and N receiving antennas arranged as a uniform linear array, the RIS has M elements, and both the users and the target have single antennas. For simplicity, we set $\mathcal{N} = \{1, 2, \cdots, N\}$, $\mathcal{M} = \{1, 2, \cdots, M\}$, and $\mathcal{K} = \{1, 2, \cdots, K\}$. Furthermore, we assume that the BS operates in full-duplex mode with the aid of advanced signal processing techniques [41], endowing the BS with perfect self-interference mitigation capabilities[1] [12]. Therein, an RIS is employed to enhance both sensing and secure communication performance. Meanwhile, the sensing target[2] attempts to eavesdrop on the confidential information between the BS and the user from the hybrid signal. Furthermore, it is posited that within the ISAC framework, communication users actively interact with the BS and the RIS. Consequently, the CSI of the communicating users is accessible through the deployment of sophisticated channel estimation techniques [42], [43]. In contrast, the identified target is ascertained by the BS during preceding target detection efforts and is delineated as a potential Eve. Therefore, the BS can obtain the CSI of Eve based on the sensing echo information. The transmission protocol can be described in the following steps: First, the CSI of users is obtained at the BS by the user sending pilot signals, and target information is obtained by analyzing the echo signals reflected from the target. Then, the BS calculates the beamforming factor of the BS and the phase shifts of the RIS using the proposed algorithm. Finally, the BS sends the optimized phase shift information to the RIS controller. Subsequently, the signal is transmitted to communicate with the user and sense the target.

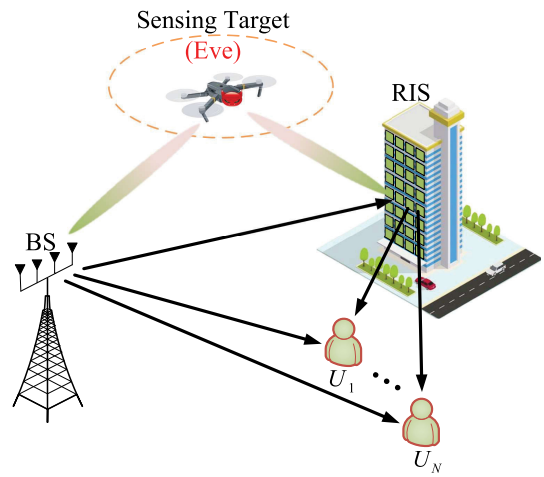

Fig. 1. System model.

A. Communications Model

The waveforms containing information and radar signals are transmitted by the BS's antennas for secure communication and sensing, and the BS's transmitted signal can be denoted as

$$\mathbf{x} = \sum_{k=1}^{K} \mathbf{w}_k s_{c,k} + \mathbf{f}_r s_r, \tag{1}$$

[1]Due to the different abilities of BS to eliminate self-interference, it is also possible that self-interference cannot be completely removed. The level of residual self-interference could impact the sensing capability at the receiving end. This is beyond the scope of this paper, and we will consider this issue in the future.

[2]Despite the single Eve focus of this paper, we can easily extend the proposed problem formulation to multiple objectives. Furthermore, the proposed algorithm can be adapted for a multi-Eve scenario with only minor modifications.

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where $\mathbf{w}_k \in \mathbb{C}^N$ and $\mathbf{f}_r \in \mathbb{C}^N$ are the matrices for communication and the dedicated radar signals, respectively. $s_{c,k}$ denotes the communication symbol of the user k and satisfies $\mathbb{E}\left\{|s_{c,k}|^2\right\} = 1$, s_r is the dedicated radar signal with $\mathbb{E}\left\{|s_r|^2\right\} = 1$. Furthermore, it is posited that communication and radar signals exhibit statistical independence and uncorrelated, i.e., $\mathbb{E}\left\{s_{c,k}s_r^\dagger\right\} = 0$. Then, the received signal at user k is represented as

$$y_k = \left(\mathbf{h}_{s,k}^\dagger + \mathbf{h}_{r,k}^\dagger \mathbf{\Phi}\mathbf{H}\right)\mathbf{x} + n_k, \tag{2}$$

where $\mathbf{h}_{s,k} \in \mathbb{C}^N$ and $\mathbf{h}_{r,k} \in \mathbb{C}^M$ are the channel vectors between the BS and RIS to user k, respectively, they are assumed to follow Rician fading are denoted as

$$\mathbf{h}_{i,k} = \zeta_{i,k}\left(\sqrt{\frac{\kappa_{i,k}}{\kappa_{i,k}+1}}\widetilde{\mathbf{h}}_{i,k}^{LoS} + \sqrt{\frac{1}{\kappa_{i,k}+1}}\widetilde{\mathbf{h}}_{i,k}^{NLoS}\right), \tag{3}$$

where $i \in \{s, r\}$, $\zeta_{i,k} = \sqrt{\beta_0 (d_{i,k})^{-\sigma_{i,k}}}$ is the distance-dependent path loss, in which β_0 is the path loss per unit distance, $d_{i,k}$ is the distance from i to k, and $\sigma_{i,k}$ is the path loss exponent. In addition, $\kappa_{i,k}$ denotes the Rician factor; $\widetilde{\mathbf{h}}_{i,k}^{\mathrm{LoS}}$ and $\widetilde{\mathbf{h}}_{i,k}^{\mathrm{NLoS}}$ are the LoS and non-LoS (NLoS) components of the channel, respectively. Similarly, $\mathbf{H} \in \mathbb{C}^{M\times N}$ is the channel between the BS and RIS, which also follows the Rician fading. Additionally, the semi-static characteristic of the BS–RIS channel permits the application of sophisticated channel estimation techniques, such as the dual time-scale channel estimation framework [43], to accurately estimate the channel parameter $\mathbf{H}$. $\mathbf{\Phi} \triangleq \mathrm{diag}\left(\mathbf{r}^\dagger\right)$ is the reflection matrix of the RIS, and $\mathbf{r}^\dagger \triangleq [r_1, r_2, \cdots r_M]$ denotes the vector of reflection coefficients with $r_m = e^{j\theta_m}, m \in \mathcal{M}$. Finally, $n_k \sim \mathcal{CN}\left(0, \sigma_k^2\right)$ means the additive white Gaussian noise (AWGN) at user k.

B. Sensing Model

Regarding the target, the transmitted signal traverses to the target via both the direct and the reflected links, subsequently being echoed back to the BS through these two links. Additionally, owing to the passive characteristics of the RIS, self-interference is circumvented, even when it operates in full duplex mode. Given the BS's exhaustive comprehension of the information waveform being transmitted, it becomes plausible to leverage communication signals for the purpose of target sensing in tandem with the dedicated radar signals. Consequently, the signal that is received at the BS, having been reflected from the target, is succinctly represented as

$$\mathbf{y}_r = \beta_t\left(\mathbf{h}_{s,e} + \mathbf{H}^\dagger\mathbf{\Phi}\mathbf{h}_{r,e}\right)\left(\mathbf{h}_{s,e}^\dagger + \mathbf{h}_{r,e}^\dagger\mathbf{\Phi}\mathbf{H}\right)\mathbf{x} + \mathbf{n}_r, \tag{4}$$

where $\beta_t \sim \mathcal{CN}\left(0, \nu^2\right)$ represents the complex-valued radar cross section (RCS) coefficient of the target. $\mathbf{n}_r \sim \mathcal{CN}\left(\mathbf{0}, \sigma_r^2\mathbf{I}_N\right)$ denotes the AWGN at the receiving end of the BS. $\mathbf{h}_{s,e} \in \mathbb{C}^N$ and $\mathbf{h}_{r,e} \in \mathbb{C}^M$ represent the channels from the BS/RIS to the target, and they are modelled as LoS channels.[3] Specifically, $\mathbf{h}_{s,e} = \zeta_{s,e}\boldsymbol{\alpha}\left(\theta_{s,e}\right)$ and $\mathbf{h}_{r,e} = \zeta_{r,e}\boldsymbol{\alpha}\left(\theta_{r,e}\right)$, where $\zeta_{s,e}$ and $\zeta_{r,e}$ are the path losses; $\theta_{s,e}$ and $\theta_{r,e}$ are the angles of arrival (DoAs) from the BS/RIS to the target. The details of the steering vector of the BS are given as

$$\boldsymbol{\alpha}\left(\theta_{i,e}\right) = \left[1, e^{-j2\pi d_i/\lambda\sin(\theta_{i,e})}, \cdots, e^{-j2\pi(N-1)d_i\lambda\sin(\theta_{i,e})}\right]^T, \tag{5}$$

where $i \in \{s, r\}$ and λ is the wavelength. d_s and d_r denote the adjacent antenna spacing and adjacent reflector spacing, respectively.

In the target detection phase, the principal objective involves acquiring distance and angular information pertaining to the target. Given the system model, our concentration is primarily on the target tracking phase; thus, the angle and distance parameters are derived from the outcomes of the preceding detection phase [36], [37], [44].[4] However, due to hardware constraints, inevitable estimation inaccuracies, and the dynamic nature of the target, there are errors in the distance and angle of the target estimated by the BS [36]. Furthermore, given its role as a potential Eve, the target does not engage in active communication with the BS for the purpose of error rectification.

Specifically, we model the distance between the BS and the target as $d_{s,e} = \overline{d}_{s,e} + \Delta d_{s,e}$, where $\overline{d}_{s,e}$ is the estimate of the distance and the bound on the uncertainty of the distance is assumed to be D_s, i.e. $|\Delta d_{s,e}| \leq D_s$. Similarly, we model the angle uncertainty as $\theta_{s,e} = \overline{\theta}_{s,e} + \Delta\theta_{s,e}$, where $\overline{\theta}_{s,e}$ is the angle estimate; the uncertainty boundary for the angle is denoted φ_s, i.e. $|\Delta\theta_{s,e}| \leq \varphi_s$. In summary, the channel of the BS-target can be rewritten as

$$\mathbf{h}_{s,e}\left(\widehat{d}_{s,e}, \widehat{\theta}_{s,e}\right) = \sqrt{\beta_0\left(\bar{d}_{s,e} + \Delta d_{s,e}\right)^{-\sigma_{s,e}}}\,\alpha\left(\bar{\theta}_{s,e} + \Delta\theta_{s,e}\right), \tag{6}$$

where $\widehat{d}_{s,e} = \bar{d}_{s,e} + \Delta d_{s,e}$ and $\widehat{\theta}_{s,e} = \bar{\theta}_{s,e} + \Delta\theta_{s,e}$. Moreover, the estimation of RIS-target channel parameters $d_{r,e}$ and $\theta_{r,e}$ at the RIS side poses significant challenges, primarily because the RIS is passive and devoid of data processing capabilities. Therefore, the RIS-target channel parameters can be estimated by leveraging the acquired sensing information and the fact that the position and angle of the RIS with respect to the BS are fixed and known. However, the uncertainties involved in estimating the target parameters by the BS can impact the accuracy of the target's parameter estimations with respect to the RIS. Similarly, the channel from the RIS to the target is remodeled as

$$\mathbf{h}_{r,e}\left(\widehat{d}_{r,e}, \widehat{\theta}_{r,e}\right) = \sqrt{\beta_0\left(\bar{d}_{r,e} + \Delta d_{r,e}\right)^{-\sigma_{r,e}}}\,\alpha\left(\bar{\theta}_{r,e} + \Delta\theta_{r,e}\right), \tag{7}$$

where $\widehat{d}_{r,e} = \bar{d}_{r,e} + \Delta d_{r,e}$, $\widehat{\theta}_{r,e} = \bar{\theta}_{r,e} + \Delta\theta_{r,e}$. $\overline{d}_{r,e}$ and $\overline{\theta}_{r,e}$ are the estimates of the distance and angle from the RIS to

[3]In a significant portion of the ISAC literature [32], sensing channels are predominantly modeled as LoS links in scenarios where the sensed target is airborne or the radio propagation environment is characterized by the absence of scatterers.

[4]In the early detection phase, the BS lacks pre-existing knowledge about the target, prompting the emission of omni-directional signals to surmise the target's parameters. Upon completion of the initial detection phase, the BS acquires estimated values of the target's parameters, notably its distance and angle. These estimates are subsequently leveraged in the tracking phase to facilitate transmit beamforming and downlink communication.

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the target, respectively. $\left|\bar{d}_{r,e}\right| \leq D_r$ and $|\Delta\theta_{r,e}| \leq \varphi_r$ are the uncertainty boundaries for the distance and angle, respectively.

III. Preformance Metrics and Problem Formation

In this section, we introduce the performance metrics for secure communication and radar sensing, deemed crucial for assessing the pertinent performance. Moreover, we assess the performance of multiuser secure communication employing standard sum secrecy rate and evaluate the sensing capability through beampattern gain. Subsequently, we formulate the optimisation problem.

A. Security Communication Metric

According to the previous model, the signal-to-interference-plus-noise ratio (SINR) of the user k is given as

$$\gamma_k = \frac{\left|\mathbf{h}_k^\dagger \mathbf{w}_k\right|^2}{\sum_{i\neq k}^K \left|\mathbf{h}_k^\dagger \mathbf{w}_i\right|^2 + \sum_{n=1}^N \left|\mathbf{h}_k^\dagger \mathbf{f}_r\right|^2 + \sigma_k^2}, \tag{8}$$

where $\mathbf{h}_k^\dagger = \left(\mathbf{h}_{s,k}^\dagger + \mathbf{r}^\dagger \mathbf{G}_k\right)$ with $\mathbf{G}_k = \operatorname{diag}\left(\mathbf{h}_{r,k}^\dagger\right)\mathbf{H}$. Similarly, the received SINR at the target eavesdropping on user k is

$$\gamma_e^k = \frac{\left|\mathbf{h}_e^\dagger \mathbf{w}_k\right|^2}{\sum_{i\neq k}^K \left|\mathbf{h}_e^\dagger \mathbf{w}_i\right|^2 + \sum_{n=1}^N \left|\mathbf{h}_e^\dagger \mathbf{f}_r\right|^2 + \sigma_e^2}, \tag{9}$$

where $\mathbf{h}_e^\dagger = \mathbf{h}_{s,e}^\dagger\left(\widehat{d}_{s,e}, \widehat{\theta}_{s,e}\right) + \mathbf{r}^\dagger \mathbf{G}_e$, and $\mathbf{G}_e = \operatorname{diag}\left(\mathbf{h}_{r,e}^\dagger\left(\widehat{d}_{r,e}, \widehat{\theta}_{r,e}\right)\right)\mathbf{H}$. Thus, it is worth noting that the CSI of the Eve included in (9) is reconstructed from imperfect sensing information. To simplify the expression, we will use $\mathbf{h}_{s,e}$ and $\mathbf{h}_{r,e}$ to represent $\mathbf{h}_{s,e}\left(\widehat{d}_{s,e}, \widehat{\theta}_{s,e}\right)$ and $\mathbf{h}_{r,e}\left(\widehat{d}_{r,e}, \widehat{\theta}_{r,e}\right)$ in the following. Then, the achievable secrecy rate for a legitimate user is defined as the difference between the achievable data rate of the legitimate receiver and that of the Eve [38], [45], [46], [47]. Based on this, we adopt worst-case sum secrecy rate as the parameter to measure the secure communication performance of the RIS-assisted ISAC system, and the details can be expressed as

$$\min_{\psi} \sum_{k=1}^K \mathcal{R}_s^k = \sum_{k=1}^K \mathcal{R}_k - \max_{\psi} \mathcal{R}_e^k \tag{10}$$

where $\mathcal{R}_k = \log_2\left(1+\gamma_k\right)$ and $\mathcal{R}_e^k = \log_2\left(1+\gamma_e^k\right)$ represent the data rates of the user k and Eve eavesdropping on user k, respectively. Moreover, $\psi = \{\theta_{s,e} \in \varphi_s, \theta_{r,e} \in \varphi_r, |\Delta d_{s,e}| \leq D_s, |\Delta d_{r,e}| \leq D_r\}$ represents the maximum possible eavesdropping rate under the uncertainties of Eve's position. This approach provides a conservative yet effective strategy to mitigate eavesdropping risk by accounting for the worst-case scenario, thereby enhancing the PLS.

B. Sensing Metric

Within the ambit of the target tracking task, in order to obtain better detection performance, a higher detection probability can be obtained by boosting the beampattern gain in the target direction. Thus, the beampattern gains, corresponding to the target, are designated as [39]

$$\mathcal{P}_{bp} = \left\{\left|\mathbf{h}_e^\dagger \mathbf{x}\right|^2\right\} = \mathbf{h}_e^\dagger\left(\sum_{k=1}^K \mathbf{w}_k \mathbf{w}_k^\dagger + \mathbf{f}_r \mathbf{f}_r^\dagger\right)\mathbf{h}_e. \tag{11}$$

C. Problem Formulation

Our goal is to jointly optimize active and passive beamforming to maximize the achievable sum secrecy rate under the constraints of guaranteeing sensing function, the communication QoS, and the total transmit power. In addition, an estimate of the target position and angle is obtained in the BS rather than an accurate one. Consequently, this leads us to formulate the following optimization problem

$$\begin{aligned}
&\underset{\{\mathbf{w}_k\}_{k=1}^K, \mathbf{f}_r, \mathbf{r}}{\text{maximize}} \sum_{k=1}^K \mathcal{R}_k - \max_{\substack{\theta_{s,e}\in\varphi_s, \theta_{r,e}\in\varphi_r, \\ |\Delta d_{s,e}|\leq D_s, |\Delta d_{r,e}|\leq D_r}} \mathcal{R}_e^k \\
&\text{subject to C1}: \mathcal{R}_k \geq R_{th}^k \\
&\qquad \text{C2}: \mathcal{P}_{bp} \geq \kappa_{th}, \\
&\qquad \text{C3}: \sum_{k=1}^K \|\mathbf{w}_k\|^2 + \|\mathbf{f}_r\|^2 \leq P_{th}, \\
&\qquad \text{C4}: |r_m| = 1, \forall m.
\end{aligned} \tag{12}$$

In order to ensure that each communication user receives a satisfying communication service, the achieved rate of the user k must satisfy the rate threshold R_{th}^k. In addition, the beampattern gain is limited by a rate threshold κ_{th}, and P_{th} denotes the total transmit power.

We note that the optimization problem in (12) is nonconvex, attributable to various factors. Specifically, the interdependency among variables coupled with fractional SINR expressions within the objective and constraint functions contributes to the nonconvex nature. At present, there is no systematic approach to solve such a problem. Therefore, in the next section, we embark on an exploration of potential solutions to this problem.

IV. Proposed Security Approximation Solution

In this section, in order to solve the optimization problem in (12), a bound on the uncertainty region of the target's CSI induced by the dual uncertainty $\Delta d_{i,e}$ and $\Delta\theta_{i,e}$ terms with $i \in \{s, r\}$ is first derived on the basis of the security approximation. Then, we transform the resulting problem into an equivalent form. Subsequently, we use an AO-based iteration approach to obtain the results.

A. Bound for Uncertainty Region of Target' CSI

In this subsection, to accomplish the design of robust security strategies, we derive a bound based on analytic geometry theory. We first consider the fading component of one entry of the channel vector $\mathbf{h}_{s,e}$ of the target, i.e., $R_n e^{-j2\pi(n-1)d_s/\lambda \sin\left(\bar{\theta}_{s,e}+\Delta\theta_{s,e}\right)}$ with $R_n =$

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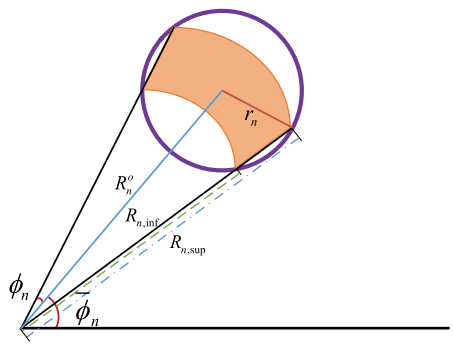

Fig. 2. Uncertainty bound based on geometric analysis.

$\sqrt{\beta_0\left(\bar{d}_{s,e}+\Delta d_{s,e}\right)^{-\sigma_{s,e}}}$. As shown in Fig. 2, for a given $R_n e^{-j2\pi(n-1)d_s/\lambda \sin\left(\bar{\theta}_{s,e}+\Delta\theta_{s,e}\right)}$ as an point with angle $\bar{\phi}_n = -2\pi\left(n-1\right)d_s/\lambda \sin\left(\bar{\theta}_{s,e}+\Delta\theta_{s,e}\right)$ and radius R_n. Then, the uncertainty in the distance makes the large-scale fading component satisfy $R_n \in [R_{n,\inf}, R_{n,\sup}]$ with $R_{n,\inf} = \sqrt{g_0\left(\bar{d}_{s,e}+D_s\right)^{-\sigma_{s,e}}}$ and $R_{n,\sup} = \sqrt{g_0\left(\bar{d}_{s,e}-D_s\right)^{-\sigma_{s,e}}}$. For an angular uncertainty $\Delta\theta_{s,e}$, the response is bounded by ϕ_n on a small-scale fading component, which is given as

$$\phi_n = \left|2\pi\left(n-1\right)d_s/\lambda\left(\sin\left(\bar{\theta}_{s,e}+\Delta\theta_{s,e}\right)-\sin\left(\bar{\theta}_{s,e}\right)\right)\right|. \tag{13}$$

As a result, the joint angle and distance uncertainty region is an orange-colored annular sector in Fig. 2. By applying analytic geometry theory [36], this region of uncertainty region is denoted as

$$S_{\mathrm{act}_n} = \phi_n\left(R_{n,\sup}^2 - R_{n,\inf}^2\right). \tag{14}$$

Nevertheless, the nonconvex nature of this uncertainty region poses a significant challenge to beamforming design, as the commonly employed methods to address uncertainty, such as the *S-procedure*, necessitate a convex uncertainty region. To address this challenge, we expand the uncertainty bound to an upper bound delineated by a circle, depicted as the purple circle in Fig. 2. According to analytic geometry theory, the radius of the region of uncertainty is given as $r_n = \frac{1}{2}\sqrt{\left(\tan\left(\phi_n\right)\left(R_{n,\inf}+R_{n,\sup}\right)\right)^2+\left(\left(R_{n,\sup}-R_{n,\inf}\right)\right)^2}$, and the amplitude R_n^o of the component can be given as $R_n^o = \frac{1}{2}\sqrt{\left(\tan\left(\phi_n\right)\left(R_{n,\inf}+R_{n,\sup}\right)\right)^2+\left(\left(R_{n,\sup}+R_{n,\inf}\right)\right)^2}$. Then, the approximated uncertainty region can be denoted as

$$S_{\mathrm{app}_n} = \pi r_n^2. \tag{15}$$

Subsequently, the precision of the boundary is quantified by calculating the ratio of the actual area to the corresponding boundary area, denoted as

$$\eta_{\mathrm{ratio}} = \frac{\sum_{n=1}^{N} S_{\mathrm{act}_n}}{\sum_{n=1}^{N} S_{\mathrm{app}_n}}. \tag{16}$$

To demonstrate the validity of the approximate boundaries, we initially plot the accuracy of these boundaries concerning distance uncertainty D_s and angular uncertainty φ_s for $d_s = 0.05m$, $\lambda = 0.3m$, $\bar{\theta}_{s,e} = \pi/3$, $\bar{d}_{s,e} = 50m$, and $N = 8$,

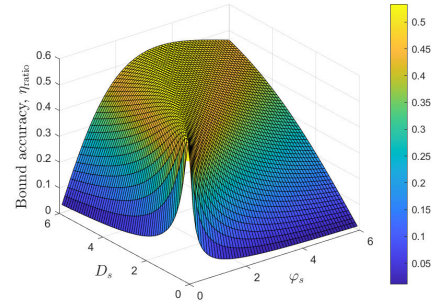

Fig. 3. Illustration the approximate boundaries.

as illustrated in Fig. 3. We observe that this bound demonstrates accuracy under conditions of high uncertainty (above 50%) in both dual parameters. However, it is acknowledged that the primary parameter influencing system performance is the channel error [48], denoted as

$$\varepsilon_{\mathrm{errors}} = \frac{\sqrt{\sum_{n=1}^{N} r_{n-1}^2}}{\sqrt{\sum_{n=1}^{N}\left(R_{n-1}^o\right)^2}}. \tag{17}$$

For this purpose, we depict the channel error as illustrated in Fig. 4. Notably, when the uncertainty of the dual parameters is minimal, the approximate boundary error for the channel may be significant, yet the actual channel error remains lower. This phenomenon is due to the fact that when the double uncertainty is small, the approximated bounds over amplify the actual bounds. Since the uncertainty remains small after the approximation, this allows the channel error to remain low and realistic. Therefore, the approximation remains safe at this point. On the contrary, when the uncertainty in the two parameters is large, the high accuracy of our approximate boundary ensures that its channeling error is comparable to the channeling error caused by the actual boundary. In conclusion, the channel boundary proposed is well-suited for the optimization framework developed in this paper. Concurrently, with the escalation of dual parameter uncertainty, the uncertainty within the channel reaches a notable level [48] (25%). Despite relatively precise boundaries, achieving effective beamforming and thus PLS proves challenging. Consequently, the system needs to restart the omnidirectional sensing detection phase until the sensing parameters are obtained within an acceptable error range, after which it transitions to the target tracking phase. Additionally, when system resources are sufficient, secure beamforming schemes assisted by isotropic artificial noise or more advanced sensing parameter estimation techniques can further enhance system security; these aspects are beyond the scope of this paper.

Therefore, $\mathbf{h}_{s,e}$ is reformulated as $\mathbf{h}_{s,e} = \overline{\mathbf{h}}_{s,e} + \Delta\mathbf{h}_{s,e}$, where $\overline{\mathbf{h}}_{s,e}$ can be given by (18), as shown at the bottom of the next page. Furthermore, the uncertainty of CSI satisfies the following bound, $\Delta\mathbf{h}_{s,e}^{\dagger}\Delta\mathbf{h}_{s,e} \leq \sum_{n=1}^{N} r_{n-1}^2 \triangleq \varepsilon_{s,e}^2$, and the continuous set containing all possible uncertainties can be represented as

$$\Omega_1 \triangleq \left\{\Delta\mathbf{h}_{s,e} \in {}^{N\times 1} : \left\|\Delta\mathbf{h}_{s,e}\right\|_2 \leq \varepsilon_{s,e}\right\}, \tag{19}$$

with its norm bounded by $\varepsilon_{s,e} > 0$.

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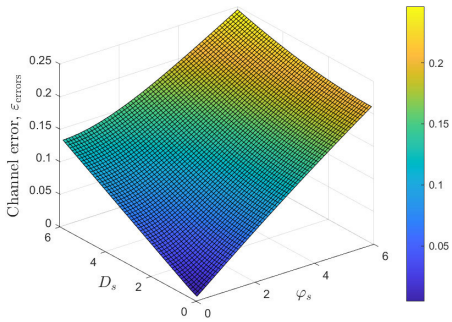

Fig. 4. Illustration the channel error.

Similarly, the same approach can be adopted for $\mathbf{h}_{r,e}$, which is further rewritten to $\mathbf{h}_{r,e} = \overline{\mathbf{h}}_{r,e} + \Delta\mathbf{h}_{r,e}$, where $\overline{\mathbf{h}}_{r,e}$ and the uncertainty $\Delta\mathbf{h}_{r,e}$ are also obtained according to the above scheme. Moreover, the continuous set containing all possible uncertainties can be denoted as follows:

$$\Omega_2 \triangleq \left\{\Delta\mathbf{h}_{r,e} \in {}^{M\times 1} : \left\|\Delta\mathbf{h}_{r,e}\right\|_2 \leq \varepsilon_{r,e}\right\}. \tag{20}$$

where the norm boundary $\varepsilon_{r,e} > 0$ is likewise obtained similarly to $\varepsilon_{s,e}$.

B. Robust Security Approximation Solution

In this subsection, we utilize the obtained target channel boundary as a lower bound to replace the original problem. First, we introduce the slack variables ∂_k and ς_k, and we can reformulate (12) as

$$\begin{aligned} &\underset{\{\mathbf{w}_k\}_{k=1}^K, \mathbf{f}_r, \mathbf{r}}{\text{maximize}} \sum\nolimits_{k=1}^K \partial_k - \varsigma_k \\ &\text{subject to C1} \sim \text{C4}, \\ &\qquad \text{C5}: \mathcal{R}_k \geq \partial_k, \\ &\qquad \text{C6}: \varsigma_k \geq \max_{\substack{\Delta\mathbf{h}_{s,e}\in\Omega_1,\\ \Delta\mathbf{h}_{r,e}\in\Omega_2,}} \mathcal{R}_e^k. \end{aligned} \tag{21}$$

Next, we deal with the variational constraint C5 on the objective function. For simplicity, we first introduce the exponential auxiliary variables $\{\vartheta_k\}_{k=1}^K$ and $\{\rho_k\}_{k=1}^K$ fulfilling $\partial_k = (\vartheta_k + \rho_k)\log_2 e$, which also satisfy

$$\sum\nolimits_{i=1}^K \left|\left(\mathbf{h}_{s,k}^\dagger + \mathbf{r}^\dagger\mathbf{G}_k\right)\mathbf{w}_i\right|^2 + \sigma_k^2 \geq e^{\vartheta_k}, \tag{22}$$

and

$$\sum\nolimits_{i\neq k}^K \left|\left(\mathbf{h}_{s,k}^\dagger + \mathbf{r}^\dagger\mathbf{G}_k\right)\mathbf{w}_i\right|^2 + \sigma_k^2 \leq e^{\rho_k}. \tag{23}$$

First, for a complex variable x, the following first-order Taylor inequality holds,

$$|x|^2 \geq 2\text{Re}\left\{x^{(t),*}x\right\} - x^{(t),*}x^{(t)}. \tag{24}$$

Thus, according to (24), (22) can be rewritten as (25), shown at the bottom of the next page. Again, the first-order Taylor expansion is employed to reconstruct the form of (23). We make these constraints convexly approximated at $\{\widetilde{\rho}_k\}_{k=1}^K$, and by the first-order Taylor expansions, (23) is approximated as

$$\sum\nolimits_{i\neq k}^K \left|\left(\mathbf{h}_{s,k}^\dagger + \mathbf{r}^\dagger\mathbf{G}_k\right)\mathbf{w}_i\right|^2 + \sigma_k^2 \leq e^{\widetilde{\rho}_k}\left(1 + \rho_k - \widetilde{\rho}_k\right). \tag{26}$$

In addition, in order to eliminate the quadratic constraints, the SOC constraint transformation, i.e., $\xi^2 \leq \eta\chi\,(\eta \geq 0, \chi \geq 0) \Rightarrow \left\|[2\xi, \eta - \chi]^\dagger\right\| \leq \eta + \chi$, is adopted. Thus, (26) is further transformed into (27), shown at the bottom of the next page, where $\Gamma = e^{\widetilde{\rho}_k}\left(1 + \rho_k - \widetilde{\rho}_k\right)$.

After relaxing the target uncerdtainty bound, the complexity of the target's channel capacity expression is significantly changed, but the semi-infinite inequality constraints in C6 still make it intractable. To this end, we first introduce relaxation variables $\{\mu_k\}_{k=1}^K$ and $\{\nu_k\}_{k=1}^K$, the fractions in C6 are rewritten as

$$\left|\left(\mathbf{h}_{s,e}^\dagger + \mathbf{r}^\dagger\mathbf{G}_e\right)\mathbf{w}_k\right|^2 \leq \mu_k, \tag{28}$$

and

$$\sum\nolimits_{i\neq k}^K \left|\left(\mathbf{h}_{s,e}^\dagger + \mathbf{r}^\dagger\mathbf{G}_e\right)\mathbf{w}_i\right|^2 + \left|\left(\mathbf{h}_{s,e}^\dagger + \mathbf{r}^\dagger\mathbf{G}_e\right)\mathbf{f}_r\right|^2 + \sigma_k^2 \geq \nu_k. \tag{29}$$

Furthermore, by introducing $\{\chi_k\}_{k=1}^K$, C6 is further reconstructed as

$$\mu_k/\nu_k \leq \chi_k, \tag{30}$$

and

$$\ln\left(1 + \chi_k\right) \leq \dot{\varsigma}_k. \tag{31}$$

where $\dot{\varsigma}_k = \ln 2\varsigma_k$.

To address the uncertainty constraints (28) and (29), we first introduce the following two lemmas about multiple complex-valued uncertainties, which will be used in later derivations.

*Lemma 1 (*General $\mathcal{S}$-Procedure *[49]):* Let $f_q(\mathbf{x}) = \mathbf{x}^H\mathbf{F}_q\mathbf{x} + 2\text{Re}\left(\mathbf{g}_q^H\mathbf{x}\right) + h_q, q = 1, 2, \cdots, Q$ where $\mathbf{x} \in \mathbb{C}^{L\times 1}$, $\mathbf{F}_q = \mathbf{F}_q^\dagger$, $\mathbf{g}_q \in \mathbb{C}^{L\times 1}$, and $h_q \in \mathbb{R}$. We can get the expression $\{f_q(\mathbf{x}) \geq 0\}_{q=2}^Q \Rightarrow f_1(\mathbf{x}) \geq 0$ hold and only if there exists $\lambda_q \geq 0, \forall q$ such that

$$\begin{bmatrix} \mathbf{F}_1 & \mathbf{g}_1 \\ \mathbf{g}_1^H & h_1 \end{bmatrix} - \sum_{q=2}^Q \lambda_q \begin{bmatrix} \mathbf{F}_q & \mathbf{g}_q \\ \mathbf{g}_q^H & h_q \end{bmatrix} \succeq 0. \tag{32}$$

*Lemma 2 (*General Sign-Definiteness *[50]):* Let $\mathbf{F} \succeq \sum_{q=1}^Q \left(\mathbf{A}_q^\dagger\mathbf{X}_q\mathbf{B}_q + \mathbf{B}_q^\dagger\mathbf{X}_q^\dagger\mathbf{A}_q\right)$ with $\|\mathbf{X}_q\|_F \leq \varepsilon_q$ and $\mathbf{X}_q = \mathbf{X}_q^\dagger$. We can obtain the condition

$$\overline{\mathbf{h}}_{s,e} = \left[R_0^o, R_1^o e^{-j\frac{2\pi d_i}{\lambda}\sin\left(\bar{\theta}_{s,e}\right)}, \cdots, R_{(N-1)}^o e^{-j\frac{2\pi d_i}{\lambda}(N-1)\sin\left(\bar{\theta}_{s,e}\right)}\right]^T \tag{18}$$

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$\left\{\|\mathbf{X}_q\|_F \le \varepsilon_q\right\}_{q=1}^{Q} \Rightarrow \mathbf{F} \succeq \sum_{q=1}^{Q}\left(\mathbf{A}_q^\dagger \mathbf{X}_q \mathbf{B}_q + \mathbf{B}_q^\dagger \mathbf{X}_q^\dagger \mathbf{A}_q\right)$ holds if and only if there exists $\lambda_q \ge 0, \forall q$ such that

$$\begin{bmatrix} \mathbf{F} - \sum_{q=1}^{Q} \widetilde{\lambda}_q \mathbf{B}_q^\dagger \mathbf{B}_q & -\varepsilon_1 \mathbf{A}_1^\dagger & \cdots & -\varepsilon_Q \mathbf{A}_Q^\dagger \\ -\varepsilon_1 \mathbf{A}_1 & \widetilde{\lambda}_1 \mathbf{I} & \cdots & \mathbf{0} \\ \vdots & \vdots & \ddots & \vdots \\ -\varepsilon_Q \mathbf{A}_Q & \mathbf{0} & \cdots & \widetilde{\lambda}_Q \mathbf{I} \end{bmatrix} \succeq 0. \quad (33)$$

To deal with constraint (28), we first transform it equivalently to LMI according to the Schur's complement, i.e.,

$$\begin{bmatrix} \mu_k & \left(\mathbf{h}_{s,e}^\dagger + \mathbf{r}^\dagger \mathbf{G}_e\right) \mathbf{w}_k \\ \mathbf{w}_k^\dagger \left(\mathbf{h}_{s,e} + \mathbf{r}\mathbf{G}_e^\dagger\right) & 1 \end{bmatrix} \succeq \mathbf{0}_2. \quad (34)$$

Then, we substitute $\mathbf{h}_{s,e} = \overline{\mathbf{h}}_{s,e} + \Delta\mathbf{h}_{s,e}$ and $\mathbf{h}_{r,e} = \overline{\mathbf{h}}_{r,e} + \Delta\mathbf{h}_{r,e}$ into (34) to obtain the following inequality

$$\begin{bmatrix} \mu_k & \widetilde{h}_k^\dagger \\ \widetilde{h}_k & 1 \end{bmatrix} + \begin{bmatrix} \Delta\mathbf{h}_{s,e}^\dagger \\ 0 \end{bmatrix} [0 \ \mathbf{w}_k] + \begin{bmatrix} 0 \\ \mathbf{w}_k^\dagger \end{bmatrix} [\Delta\mathbf{h}_{s,e} \ 0] + \begin{bmatrix} \Delta\mathbf{h}_{r,e}^\dagger \\ 0 \end{bmatrix} [0 \ \boldsymbol{\Upsilon}_k] + \begin{bmatrix} 0 \\ \boldsymbol{\Upsilon}_k^\dagger \end{bmatrix} [\Delta\mathbf{h}_{r,e} \ 0] \succeq \mathbf{0}_2, \quad (35)$$

where $\widetilde{h}_k = \left(\mathbf{w}_k^\dagger \overline{\mathbf{h}}_{s,e} + \mathbf{w}_k^\dagger \mathbf{H}^\dagger \mathrm{diag}\left(\overline{\mathbf{h}}_{r,e}\right) \mathbf{r}\right)$ and $\boldsymbol{\Upsilon}_k = \mathrm{diag}\left(\mathbf{r}^\dagger\right) \mathbf{H}\mathbf{w}_k$. Furthermore, by combining the channel uncertainty bounds (19) and (20) and introducing the slack variables $\{\varpi_{s,k}, \varpi_{s,k}\}_{k=1}^{K}$, (35) is further equivalently transformed via *Lemma 2* into the following equivalent LMI,

$$\text{C6a}: \begin{bmatrix} \varpi_k & \widetilde{h}_k^\dagger & \mathbf{0}_{1\times N} & \mathbf{0}_{1\times N} \\ \widetilde{h}_k & 1 & -\varepsilon_{r,e}\boldsymbol{\Upsilon}_k^\dagger & -\varepsilon_{s,e}\mathbf{w}_k^\dagger \\ \mathbf{0}_{N\times 1} & -\varepsilon_{r,e}\boldsymbol{\Upsilon}_k & \varpi_{r,k}\mathrm{I}_N & \mathbf{0}_{N\times N} \\ \mathbf{0}_{N\times 1} & -\varepsilon_{s,e}\mathbf{w}_k & \mathbf{0}_{N\times N} & \varpi_{s,k}\mathrm{I}_N \end{bmatrix} \succeq \mathbf{0}_{2N+2}, \quad (36)$$

where $\varpi_k = \left(e^{\varsigma_k \ln 2} - 1\right)\nu_k - M\varpi_{r,k} - \varpi_{s,k}$.

Then, according to (24), we can obtain a lower bound for $\left|\left(\mathbf{h}_{s,e}^\dagger + \mathbf{r}^\dagger \mathbf{G}_e\right)\mathbf{w}_i\right|^2$ in (29) by the first-order Taylor expansion, i.e.,

$$\begin{aligned} &\left|\left(\mathbf{h}_{s,e}^\dagger + \mathbf{r}^\dagger \mathbf{G}_e\right)\mathbf{w}_i\right|^2 \\ &\ge \left(\mathbf{h}_{s,e}^\dagger + \mathbf{r}^{(t),\dagger}\mathbf{G}_e\right)\mathbf{w}_i^{(t)}\mathbf{w}_i^\dagger\left(\mathbf{h}_{s,e} + \mathbf{G}_e^\dagger \mathbf{r}\right) \\ &+ \left(\mathbf{h}_{s,e}^\dagger + \mathbf{r}^\dagger \mathbf{G}_e\right)\mathbf{w}_i\mathbf{w}_i^{(t),\dagger}\left(\mathbf{h}_{s,e} + \mathbf{G}_e^\dagger \mathbf{r}^{(t)}\right) \\ &- \left(\mathbf{h}_{s,e}^\dagger + \mathbf{r}^{(t),\dagger}\mathbf{G}_e\right)\mathbf{w}_i^{(t)}\mathbf{w}_i^{(t),\dagger}\left(\mathbf{h}_{s,e} + \mathbf{G}_e^\dagger \mathbf{r}^{(t)}\right). \end{aligned} \quad (37)$$

In particular, the linear approximation in (37) is given in the following lemma.

Lemma 3: Set $\mathbf{w}_i^{(t)}$ and $\mathbf{r}^{(t)}$ as the optimal solutions obtained at the t-th iteration, and by inserting $\mathbf{h}_{s,e} = \overline{\mathbf{h}}_{s,e} + \Delta\mathbf{h}_{s,e}$ and $\mathbf{h}_{r,e} = \overline{\mathbf{h}}_{r,e} + \Delta\mathbf{h}_{r,e}$ into (37), the lower bound approximation function of $\left|\left(\mathbf{h}_{s,e}^\dagger + \mathbf{r}^\dagger \mathbf{G}_e\right)\mathbf{w}_i\right|^2$ can be obtained by

$$\mathbf{x}_e^H \mathbf{A}_i \mathbf{x}_e + 2\mathfrak{Re}\left\{\widetilde{\mathbf{a}}_i^\dagger \mathbf{x}_e\right\} + \widetilde{a}_i, \quad (38)$$

where

$$\begin{aligned} \mathbf{x}_e &= \left[\Delta\mathbf{h}_{s,e}^\dagger \ \ vec^\dagger\left(\mathrm{diag}\left(\Delta\mathbf{h}_{r,e}\right)\right)\right]^\dagger, \\ \mathbf{A}_i &= \mathbf{F}_i + \mathbf{F}_i^\dagger - \mathbf{Z}_i, \\ \widetilde{\mathbf{a}}_i &= \widetilde{\mathbf{b}}_{1,i} + \widetilde{\mathbf{b}}_{2,i} - \widetilde{\mathbf{d}}_i, \\ \widetilde{a}_i &= 2\mathfrak{Re}\left\{\widetilde{c}_i\right\} - \widetilde{e}_i, \\ \mathbf{F}_i &= \begin{bmatrix} \mathbf{w}_i^{(t)} \\ \mathbf{H}\mathbf{w}_i^{(t)} \otimes \mathbf{r}^{(t),*} \end{bmatrix} \left[\mathbf{w}_i^\dagger \ \ \mathbf{w}_i^\dagger \mathbf{H}^\dagger \otimes \mathbf{r}^T\right], \\ \mathbf{Z}_i &= \begin{bmatrix} \mathbf{w}_i^{(t)} \\ \mathbf{H}\mathbf{w}_i^{(t)} \otimes \mathbf{r}^{(t),*} \end{bmatrix} \left[\mathbf{w}_i^{(t),\dagger} \ \ \mathbf{w}_i^{(t),\dagger}\mathbf{H}^\dagger \otimes \mathbf{r}^{(t),T}\right], \\ \widetilde{\mathbf{b}}_{1,i} &= \begin{bmatrix} \mathbf{w}_i\mathbf{w}_i^{(t),\dagger}\left(\overline{\mathbf{h}}_{s,e} + \overline{\mathbf{G}}_e^\dagger \mathbf{r}^{(t)}\right) \\ vec^*\left(\mathbf{r}\left(\overline{\mathbf{h}}_{s,e}^\dagger + \mathbf{r}^{(t),\dagger}\overline{\mathbf{G}}_e\right)\mathbf{w}_i^{(t)}\mathbf{w}_i^\dagger\mathbf{H}^\dagger\right) \end{bmatrix}, \\ \widetilde{\mathbf{b}}_{2,i} &= \begin{bmatrix} \mathbf{w}_i^{(t)}\mathbf{w}_i^\dagger\left(\overline{\mathbf{h}}_{s,e} + \overline{\mathbf{G}}_e^\dagger \mathbf{r}\right) \\ vec^*\left(\mathbf{r}^{(t)}\left(\overline{\mathbf{h}}_{s,e}^\dagger + \mathbf{r}^\dagger\overline{\mathbf{G}}_e\right)\mathbf{w}_i\mathbf{w}_i^{(t),\dagger}\mathbf{H}^\dagger\right) \end{bmatrix}, \\ \widetilde{\mathbf{d}}_i &= \begin{bmatrix} \mathbf{w}_i^{(t)}\mathbf{w}_i^{(t),\dagger}\left(\overline{\mathbf{h}}_{s,e} + \overline{\mathbf{G}}_e^\dagger \mathbf{r}^{(t)}\right) \\ vec^*\left(\mathbf{r}^{(t)}\left(\overline{\mathbf{h}}_{s,e}^\dagger + \mathbf{r}^{(t),\dagger}\overline{\mathbf{G}}_e\right)\mathbf{w}_i^{(t)}\mathbf{w}_i^{(t),\dagger}\mathbf{H}^\dagger\right) \end{bmatrix}, \\ \widetilde{c}_i &= \left(\overline{\mathbf{h}}_{s,e}^\dagger + \mathbf{r}^{(t),\dagger}\overline{\mathbf{G}}_e\right)\mathbf{w}_i^{(t)}\mathbf{w}_i^\dagger\left(\overline{\mathbf{h}}_{s,e} + \overline{\mathbf{G}}_e^\dagger \mathbf{r}\right), \\ \widetilde{e}_i &= \left(\overline{\mathbf{h}}_{s,e}^\dagger + \mathbf{r}^{(t),\dagger}\overline{\mathbf{G}}_e\right)\mathbf{w}_i^{(t)}\mathbf{w}_i^{(t),\dagger}\left(\overline{\mathbf{h}}_{s,e} + \overline{\mathbf{G}}_e^\dagger \mathbf{r}\right). \end{aligned}$$

Proof: Please refer to Appendix A.

Furthermore, we obtain the lower bound approximation function of (29) as

$$\mathbf{x}_e^H \mathbf{A}_{-k} \mathbf{x}_e + 2\mathfrak{Re}\left\{\widehat{\mathbf{a}}_{-k}^\dagger \mathbf{x}_e\right\} + \widehat{a}_{-k} + \sigma_k^2 \ge \nu_k, \quad (39)$$

in which $\mathbf{A}_{-k} = \sum_{i\ne k}^{K+1}\mathbf{A}_i$, $\widehat{\mathbf{a}}_{-k}^\dagger = \sum_{i\ne k}^{K+1}\widetilde{\mathbf{a}}_i^\dagger$, $\widehat{a}_{-k} = \sum_{i\ne k}^{K+1}\widetilde{a}_i$, and $\mathbf{w}_{K+1} \triangleq \mathbf{f}_r$. Then, *Lemma 1* is adopted to transform (39) to

$$\text{C6b}: \begin{bmatrix} \mathbf{A}_{-k} + \begin{bmatrix} \lambda_{s,e}\mathbf{I}_N & \mathbf{0} \\ \mathbf{0} & \lambda_{r,e}\mathbf{I}_{M^2} \end{bmatrix} & \widehat{\mathbf{a}}_{-k} \\ \widehat{\mathbf{a}}_{-k}^\dagger & \bar{\lambda}_{-k} \end{bmatrix} \ge 0, \quad (40)$$

where $\bar{\lambda}_{-k} = \widehat{a}_{-k} + \sigma_k^2 - \nu_k - \lambda_{s,e}\varepsilon_{s,e}^2 - \lambda_{r,e}\varepsilon_{r,e}^2$.

$$\text{C5a}: \sum_{i=1}^{K} 2\mathfrak{Re}\left\{\left(\mathbf{h}_{s,k}^\dagger + \mathbf{r}^{(n)}\mathbf{G}_k\right)\mathbf{w}_i^{(n)}\mathbf{w}_i^\dagger\left(\mathbf{h}_{s,k} + \mathbf{G}_k^\dagger \mathbf{r}\right)\right\} - \left(\mathbf{h}_{s,k}^\dagger + \mathbf{r}^{(n)}\mathbf{G}_k\right)\mathbf{w}_i^{(n)}\mathbf{w}_i^{(n),\dagger}\left(\mathbf{h}_{s,k} + \mathbf{G}_k^\dagger \mathbf{r}^{(n)}\right) + \sigma_k^2 \ge e^{\vartheta_k}. \quad (25)$$

$$\text{C5b}: \left\|\left[2\left(\mathbf{h}_{s,K}^\dagger + \mathbf{r}^\dagger \mathbf{G}_K\right)\mathbf{w}_K, \cdots, 2\left(\mathbf{h}_{s,i}^\dagger + \mathbf{r}^\dagger \mathbf{G}_i\right)\mathbf{w}_i, \cdots, 2\left(\mathbf{h}_{s,1}^\dagger + \mathbf{r}^\dagger \mathbf{G}_1\right)\mathbf{w}_1, 2\sigma_k, \Gamma - 1\right]^\dagger\right\| \le \Gamma + 1, i \ne k. \quad (27)$$

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Moreover, the SCA-based iterative scheme is utilized to deal with (30) and (31). First, the arithmetic geometric mean inequality, i.e., $2ab \leq \lambda a^2 + b^2/\lambda$, is used for (30). Thus, the form at iteration t is given as

$$\text{C6c}: \dot{\nu}_k \geq (\nu_k)^{-1}, \tag{41}$$

and

$$\text{C6d}: \frac{1}{2}\left(\lambda_k^t \mu_k^2 + \dot{\nu}_k^2/\lambda_k^t\right) \leq \chi_k, \tag{42}$$

where $\lambda_k^t = \mu_k^{t-1}/\dot{\nu}_k^{t-1}$. Similarly, the first-order Taylor expansion is used for (31), which is rewritten as

$$\text{C6e}: \ln\left(1+\hbar_k^t\right) + \left(\chi_k - \hbar_k^t\right)/\left(1+\hbar_k^t\right) \leq \dot{\varsigma}_k, \tag{43}$$

where $\hbar_k^t = \chi_k^{t-1}$.

After the above transformation, the objective function of (12) is converted into a series of tractable constraints, and the above method is also applied to constraints C1-C4. Thus, C1 can be easily converted to

$$\text{C1a}: \partial_k \geq R_{th}^k, k \in \mathcal{K}. \tag{44}$$

For C2, the process and results of (29) can be adopted. Then, we can obtain the lower bound of C2 as

$$\mathbf{x}_e^H \mathbf{\Sigma} \mathbf{x}_e + 2\mathfrak{Re}\left\{\mathbf{f}^\dagger \mathbf{x}_e\right\} + f \geq \kappa_{th}, \tag{45}$$

where $\mathbf{\Sigma} = \sum_{i=k}^{K+1} \mathbf{A}_i$, $\mathbf{f}^\dagger = \sum_{i=k}^{K+1} \widetilde{\mathbf{a}}_i^\dagger$, $f = \sum_{i=k}^{K+1} \widetilde{a}_i$. Again, *Lemma 1* is used to convert (45) into

$$\text{C2a}: \begin{bmatrix} \mathbf{\Sigma} + \begin{bmatrix} \widehat{\lambda}_{s,e}\mathbf{I}_N & \mathbf{0} \\ \mathbf{0} & \widehat{\lambda}_{r,e}\mathbf{I}_{M^2} \end{bmatrix} & \mathbf{f} \\ \mathbf{f}^\dagger & \widehat{\lambda} \end{bmatrix} \geq 0, \tag{46}$$

where $\widehat{\lambda} = f - \kappa_{th} - \widehat{\lambda}_{s,e}\varepsilon_{s,e}^2 - \widehat{\lambda}_{r,e}\varepsilon_{r,e}^2$.

Then, C3 can be rewritten as

$$\text{C3a}: \left\|[\mathbf{f}_r, \mathbf{w}_K, \cdots, \mathbf{w}_1]^\dagger\right\| \leq \sqrt{P_{th}}. \tag{47}$$

In summary, the secrecy rate problem (12) under the double error model can be transformed into the following optimization problem.

$$\begin{aligned} &\underset{\substack{\{\mathbf{w}_k,\vartheta_k,\rho_k,\mu_k,\nu_k,\chi_k,\dot{\varsigma}_k,\varpi_{s,k},\varpi_{s,k}\}_{k=1}^K, \\ \{\mathbf{f}_r,\mathbf{r},\lambda_{s,e},\lambda_{r,e},\hat{\lambda}_{s,e},\hat{\lambda}_{r,e}\}}}{\text{maximize}} \sum\nolimits_{k=1}^K \partial_k - \varsigma_k \\ &\text{subject to C1a, C2a, C3a, C4, C5a} \sim \text{b, C6a} \sim \text{e}. \end{aligned} \tag{48}$$

To address the persistent coupling issue within the system, an AO-based algorithm is proposed to iteratively optimize the BS beamforming vector and the RIS phase shift until convergence is achieved. The detailed procedure is outlined below.

1) Optimizing BS Beamforming Vectors: Given the RIS phase shifts, the subproblem is given by

$$\begin{aligned} &\underset{\substack{\{\mathbf{w}_k,\vartheta_k,\rho_k,\mu_k,\nu_k,\chi_k,\dot{\varsigma}_k,\varpi_{s,k},\varpi_{s,k}\}_{k=1}^K, \\ \{\mathbf{f}_r,\lambda_{s,e},\lambda_{r,e},\hat{\lambda}_{s,e},\hat{\lambda}_{r,e}\}}}{\text{maximize}} \sum\nolimits_{k=1}^K \partial_k - \varsigma_k \\ &\text{subject to C1a, C2a, C3a, C5a} \sim \text{b, C6a} \sim \text{e}. \end{aligned} \tag{49}$$

Clearly, problem (49) constitutes a Semidefinite Program (SDP) that can be efficiently resolved using a well-established toolbox such as CVX.

2) Optimizing RIS Phase Shifts: After obtaining the beamforming vectors for the BS, we optimize the RIS phase shift by

$$\begin{aligned} &\underset{\substack{\{\vartheta_k,\rho_k,\mu_k,\nu_k,\chi_k,\dot{\varsigma}_k,\varpi_{s,k},\varpi_{s,k}\}_{k=1}^K, \\ \{\mathbf{r},\lambda_{s,e},\lambda_{r,e},\hat{\lambda}_{s,e},\hat{\lambda}_{r,e}\}}}{\text{maximize}} \sum\nolimits_{k=1}^K \partial_k - \varsigma_k \\ &\text{subject to C1a, C2a, C5a} \sim \text{b, C6a} \sim \text{e}, \\ &\text{C4a}: 0 \leq |r_m| \leq 1, m \in \mathcal{M}. \end{aligned} \tag{50}$$

Similarly, (50) can be solved efficiently by CVX. Then, the phase shift on the incident signal by the m-th element of the RIS can be calculated as

$$\theta_m = \arctan \frac{\mathfrak{Im}\left(r_m\right)}{\mathfrak{Re}\left(r_m\right)}, \theta_m\left[0, 2\pi\right). \tag{51}$$

which is used to determine the optimal phase shifts for each RIS element. (51) ensures that each RIS phase shift satisfies the unit magnitude constraint, which is essential for practical passive RIS implementations.

3) AO-Based Algorithm: The original problem (12) has been transformed into two convex subproblems, namely (49) and (50). To streamline the solution of the complex optimization problem and develop readily implementable algorithms, we propose an iterative AO-based algorithm to solve them, which can be summarized in Algorithm 1. Additionally, it is worth noting that the optimization order of the subproblems is not strictly fixed.

Algorithm 1 AO Algorithm for $\{\mathbf{w}_k\}_{k=1}^K$, $\mathbf{f}_r$, and $\mathbf{r}$

1: Initialize the accuracy thresholds σ, maximum number of iterations $iter_{\max}$, initial iterations $t = 0$, feasible points $\{\mathbf{w}_k^0\}_{k=1}^K$, $\mathbf{f}_r^0$, and $\mathbf{r}^0$.
2: **repeat**
3: Given $\mathbf{r}^t$, Update $\{\mathbf{w}_k^{t+1}\}_{k=1}^K$ and $\mathbf{f}_r^{t+1}$ by solving problem (49).
4: Obtained $\{\mathbf{w}_k^{t+1}\}_{k=1}^K$ and $\mathbf{f}_r^{t+1}$, Update $\mathbf{r}^{t+1}$ by solving problem (50).
5: $t = t + 1$.
6: **until** the increase in the objective value is below σ, or the maximum number of iterations $iter_{\max}$ is reached.
7: **return** $\{\mathbf{w}_k\}_{k=1}^K$, $\mathbf{f}_r$, and $\mathbf{r}$.

C. Convergence and Complexity Analysis

1) Convergence Analysis: For given $\{\mathbf{w}_k^t, \mathbf{f}_r^t, \mathbf{r}^t\}$, the results $\{\mathbf{w}_k^{t+1}, \mathbf{f}_r^{t+1}, \mathbf{r}^t\}$ are obtained for $(t+1)$-th by solving (49), which is locally optimal and the objective value is non-decreasing. Clearly, $\{\mathbf{w}_k^{t+1}, \mathbf{f}_r^{t+1}, \mathbf{r}^t\}$ is a lower bound for the optimization problem (50), a non-decreasing results $\{\mathbf{w}_k^{t+1}, \mathbf{f}_r^{t+1}, \mathbf{r}^{t+1}\}$ can also be obtained by solving (50). Furthermore, the objective function values of (49) and (50) consistently maintain or increase after each iteration and possess a finite upper limit, thereby ensuring the convergence of Algorithm 1.

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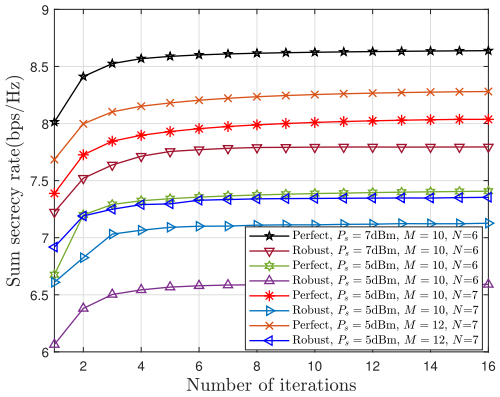


Fig. 5. Convergence performance of the objective value of Algorithm 1.

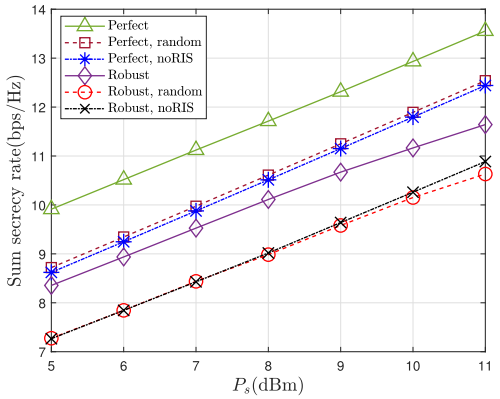


Fig. 6. Sum secrecy rate versus transmit power P_s.

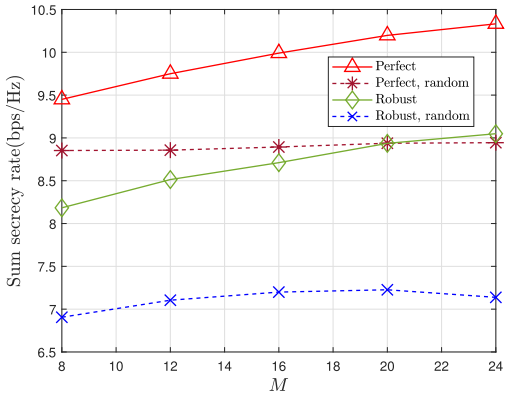


Fig. 7. Sum secrecy rate versus the Number of RIS Elements M (P_s = 5dBm).

2) Complexity Analysis: The principal computational complexity of Algorithm 1 is attributed to solving (49) and (50). According to [51], for (49), the number of iterations is $\sqrt{\Pi_w} = \sqrt{K\left(M^2+3N+10\right)}$, and the complexity of each iteration is $\Psi_w = n_w K\left(N+M^2+1\right)^3 + 8n_w K(N+1)^3 + 5n_w K + n_w(KN+N+1)^2 + n_w\left(K-1\right)\left(K+2\right)^2 + n_w^2 K\left(N+M^2+1\right)^2 + 4n_w^2 K(N+1)^2 + 5n_w^2 K + n_w^3$ where $n_w = \mathcal{O}(NK)$. Therefore, the complexity of (49) is $\sqrt{\Pi_w}\Psi_w$. Similarly, for (50), the complexity of (50) is $\sqrt{\Pi_r}\Psi_r$, in which $\sqrt{\Pi_r} = \sqrt{K\left(M^2+3N+10\right)+M-2}$, $\Psi_r = n_r K\left(N+M^2+1\right)^3 + 8n_r K(N+1)^3 + 5n_r K + n_r M^3 + n_r\left(K-1\right)\left(K+2\right)^2 + n_r^2 K\left(N+M^2+1\right)^2 + 4n_r^2 K(N+1)^2 + 5n_r^2 K + n_r^2 K M^2 + n_r^3$, and $n_r = \mathcal{O}(8K+M)$.

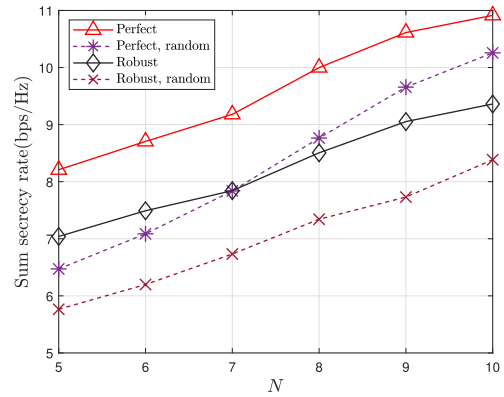
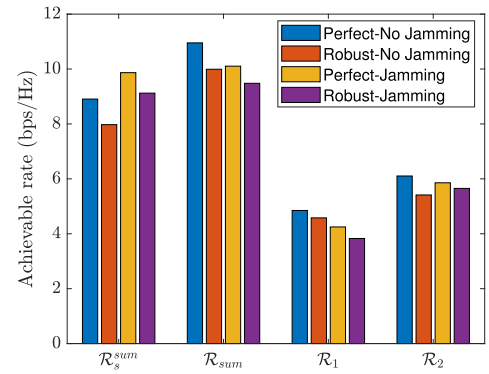
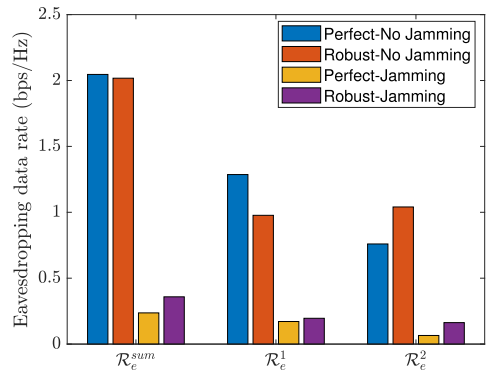


Fig. 8. Sum secrecy rate versus the Number of antennas N (P_s = 5dBm).



(a)



(b)

Fig. 9. The effect of jamming on the transmission rate: (a) The data rates of users. (b) The eavesdropping data rates of Eve (P_s = 5dBm).

V. Simulation Results

In this section, we conduct a numerical evaluation of the proposed the robust PLS scheme for RIS-assisted ISAC systems. Specifically, we set $N = 8$, $K = 2$, $M = 16$, $\sigma_k^2 = \sigma_e^2 = -80$dBm, $\nu^2 = 1$. Regarding the distance, we set the distances of the BS-RIS, the RIS-target, and the RIS-user links as $d_{s,r} = 30m$, $d_{r,e} = 8m$, and $d_{r,1} = d_{r,2} = 6m$, respectively. For the angles, we set the orientation of the RIS to the BS as $\theta_{s,r} = \pi/4$, the orientation of the BS to the RIS as $\theta_{r,s} = -\pi/4$, and the orientation of the user to the RIS as $\theta_{r,1} = \pi/7$ and $\theta_{r,2} = -\pi/7$, respectively. Then, for the remaining distance and angle parameters, they can be calculated using basic mathematical theory, e.g., $\theta_{s,e} =$

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$\text{atan}\left(\frac{d_{s,r}\sin\theta_{s,r}-d_{r,t}\cos\theta_{r,t}}{d_{s,r}\cos\theta_{s,r}-d_{r,t}\sin\theta_{r,t}}\right)$. In addition, we set the path loss indices of BS-users, BS-target, BS-RIS, RIS-target, and RIS-users links to $\sigma_{s,k} = 3.2$, $\sigma_{s,e} = 2.2$, $\sigma_{s,r} = 2$, $\sigma_{r,e} = 2$, and $\sigma_{r,k} = 2.2$, respectively. We set the Rician factor for the BS/RIS-user links and the BS-RIS link to $\kappa_{s,k} = \kappa_{s,k} = 2$dB and $\kappa_{s,r} = 4$dB, respectively. In order to evaluate the dual uncertainty in the target angle and distance parameters, we consider $D_s = 2$m, $\varphi_s = 2$degrees, $D_r = 1$m, $\varphi_r = 1$degrees. Unless otherwise specified, the key parameters are as described above.

Tag/Benchmark: The application of the proposed beamforming scheme is labeled as "Perfect" for having accurate target angle and distance estimates. To evaluate the effect of channel uncertainty due to double parameter uncertainty on performance, which is marked by "Robust". For comparative analysis, the RIS random phase shift design and the scenario without the RIS are employed as benchmarks.

A. Convergence Performance of the Algorithm 1

We present the convergence performance of the objective value of Algorithm 1 in Fig. 5. The achievable sum secrecy rate versus the number of iterations in various settings are utilized to confirm the convergence of our proposed algorithm. As can be observed, the proposed algorithm 1 converges within approximately 10-16 iterations under different parameter settings (e.g., varying transmit power P_s, number of RIS elements M, and number of antennas N). These results indicate that the proposed AO-based approach achieves convergence efficiently, which demonstrates its practical applicability.

B. Impact of Transmit Power

Then, we illustrate the total achievable secrecy rate versus transmission power in Fig. 6. Clearly, the scenario with accurate CSI consistently outperforms that with dual parameter uncertainty. Initially, focusing on "Perfect", the gains from the proposed optimization scheme compared to the baseline are substantial. Moreover, the RIS can introduce additional links, so employing random phases yields better results than that of the scenario without RIS. In cases involving uncertainty, the proposed algorithm demonstrates notable robustness and performance improvements relative to the benchmark schemes. However, it is important to note that compared to the no-RIS condition, the RIS with random phase scenario exhibits marginal gains or even a performance drop, attributable to channel uncertainty. This disparity is significantly mitigated when the proposed scheme is implemented.

C. Impact of the Number of RIS Elements

Next, the effect of the number of RIS elements versus the achievable sum secrecy rate is illustrated in Fig. 7. Initially, it is observed that with an increase in the number of elements, the achievable secrecy rate of the proposed scheme correspondingly escalates. This observation underscores the advantageous impact of augmenting the number of elements in the proposed optimization scheme. Subsequently, when the phase of the RIS remains unoptimized and is arbitrarily assigned, the enhancement in performance attributed to an increased number of elements diminishes, and the achievable sum secrecy rate for scenarios involving dual parameter uncertainty exhibits a decline when the number of RIS elements surpasses a specific threshold. In summary, the achievable secrecy rate of the proposed scheme, when equipped with perfect CSI, registers an improvement exceeding 8% relative to the scheme utilizing a random RIS phase configuration. Furthermore, under conditions of dual parameter uncertainty, this rate advances beyond 20%. Importantly, both metrics progressively increase with M.

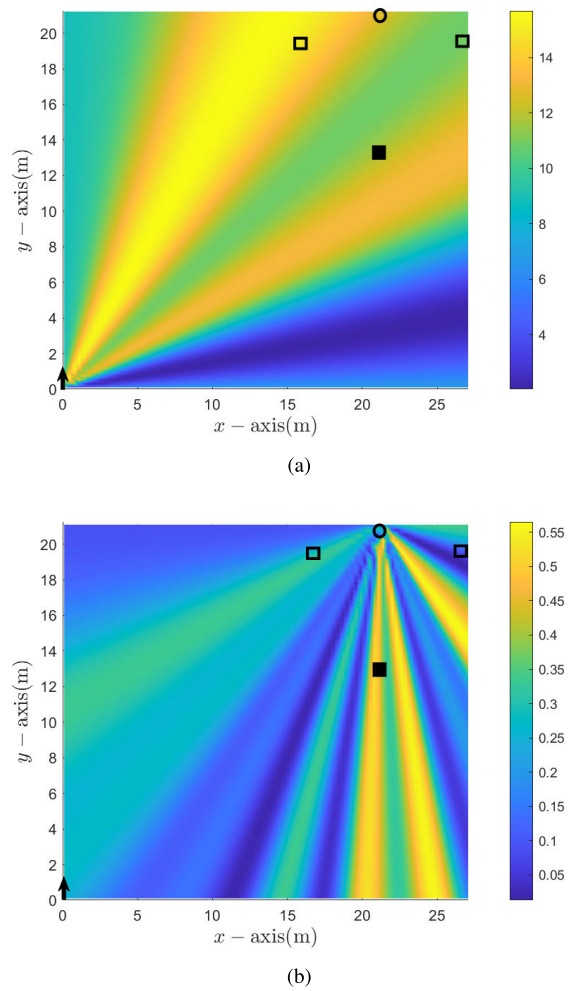

Fig. 10. Beampatterns of robust scenario: (a) beampattern of the BS. (b) beampattern of the RIS.

D. Impact of the Number of Antennas

Fig. 8 shows that the sum secrecy rate increases with the antennas of the BS. It is evident that with the increase in the number of antennas, there is a corresponding enhancement in the performance across all schemes. Additionally, in the proposed scheme utilizing perfect CSI, the performance disparity with the benchmark scheme, which employs random phases, diminishes as the antenna count escalates.

E. Impact of Jamming

To demonstrate the performance improvement of employing dedicated radar signals as jamming, we compare the user

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achievable rate and Eve's eavesdropping rate between the scenarios with jamming and without jamming. First, according to Fig. 9(a), we can observe that user 2, being closer to the BS, achieves a higher data rate than user 1 in all cases. Additionally, part of the power is allocated to interfering signals, resulting in a lower sum data rate in the presence of interference compared to its absence. Then, we see in Fig. 9(b) that the jamming significantly reduces Eve's eavesdropping data rate. Therefore, combining Fig. 9(a) and Fig. 9(b), the introduction of dedicated radar signals as jamming effectively improves the security performance. Additionally, we observed that the total rate $\mathcal{R}_{sum}$ of users without jamming is lower than that with jamming, which is attributed to the fact that the jamming signal consumes a portion of the available power.

F. Illustration of Radar Sensing Performance

We visually elucidate the dual functionalities of communication and radar sensing by depicting the beam pattern in Fig. 10, which exhibits dual parameter uncertainties. It is distinctly observable that the BS directs beams towards the region encompassing the RIS, target, and user. Concurrently, the RIS constructs several passive beams that align signals towards both the target and user.

VI. Conclusion

This paper has investigated the design of robust PLS in RIS-assisted ISAC systems with imperfect sense estimates. Specifically, we acknowledge a dual uncertainty in the sensing parameters in terms of distance and angle, and we have developed an optimization problem to maximize the sum secrecy rate. To address this, we have established a tractable boundary based on security approximations to accurately delineate the combined uncertainty of angle and distance. Subsequently, a convergence-guaranteed algorithm based on AO has been formulated employing strategies such as the $\mathcal{S}$-*procedure*, the symbolic-deterministic, the first-order Taylor expansion, and the SOC approach, and the SCA. Initial simulation results demonstrate the impact of parameter uncertainty on channel error, which escalates with increased uncertainty. Moreover, for the achievable sum secrecy rate, the proposed RIS-assisted scheme exhibits a significant performance enhancement compared to both non-RIS and randomly phased RIS configurations. For instance, with eight RIS elements, the scheme employing perfect channel conditions outperforms the random phase approach by 8%, while the dual uncertainty model records a 20% gain. Additionally, performance gains further escalate as the number of RIS elements increases. The proposed scheme could be applied to internet of things scenarios to protect against eavesdropping and unauthorized access. In future research, incorporating the environmental parameters gathered through the sensing capabilities of ISAC systems should be considered. By leveraging these parameters, such as the positions and movements of target, channel dynamics, and environmental changes, the system can reduce uncertainties and enhance the robustness of beamforming and resource allocation.

Appendix A
The Proof of Lemma 3

Substituting $\mathbf{h}_{s,e} = \overline{\mathbf{h}}_{s,e} + \Delta\mathbf{h}_{s,e}$ and $\mathbf{h}_{r,e} = \overline{\mathbf{h}}_{r,e} + \Delta\mathbf{h}_{r,e}$ to (37), we have

$$\begin{aligned}
&\left|\left(\mathbf{h}_{s,e}^{\dagger} + \mathbf{r}^{\dagger}\mathbf{G}_e\right)\mathbf{w}_i\right|^2 \\
&\geq \left(\left(\overline{\mathbf{h}}_{s,e}^{\dagger}+\Delta\mathbf{h}_{s,e}^{\dagger}\right)+\mathbf{r}^{(t),\dagger}\mathrm{diag}\left(\overline{\mathbf{h}}_{r,e}^{\dagger}+\Delta\mathbf{h}_{r,e}^{\dagger}\right)\mathbf{H}\right)\mathbf{w}_i^{(t)} \\
&\quad\times \mathbf{w}_i^{\dagger}\left(\left(\overline{\mathbf{h}}_{s,e}+\Delta\mathbf{h}_{s,e}\right)+\mathbf{H}^{\dagger}\mathrm{diag}\left(\overline{\mathbf{h}}_{r,e}+\Delta\mathbf{h}_{r,e}\right)\mathbf{r}\right) \\
&\quad+\left(\left(\overline{\mathbf{h}}_{s,e}^{\dagger}+\Delta\mathbf{h}_{s,e}^{\dagger}\right)+\mathbf{r}^{\dagger}\mathrm{diag}\left(\overline{\mathbf{h}}_{r,e}^{\dagger}+\Delta\mathbf{h}_{r,e}^{\dagger}\right)\right)\mathbf{w}_i \\
&\quad\times \mathbf{w}_i^{(t),\dagger}\left(\left(\overline{\mathbf{h}}_{s,e}+\Delta\mathbf{h}_{s,e}\right)+\mathrm{diag}\left(\overline{\mathbf{h}}_{r,e}+\Delta\mathbf{h}_{r,e}\right)\mathbf{r}^{(t)}\right) \\
&\quad-\left(\left(\overline{\mathbf{h}}_{s,e}^{\dagger}+\Delta\mathbf{h}_{s,e}^{\dagger}\right)+\mathbf{r}^{(t),\dagger}\mathrm{diag}\left(\overline{\mathbf{h}}_{r,e}^{\dagger}+\Delta\mathbf{h}_{r,e}^{\dagger}\right)\mathbf{H}\right)\mathbf{w}_i^{(t)} \\
&\quad\times \mathbf{w}_i^{(t),\dagger}\left(\left(\overline{\mathbf{h}}_{s,e}+\Delta\mathbf{h}_{s,e}\right)+\mathbf{H}^{\dagger}\mathrm{diag}\left(\overline{\mathbf{h}}_{r,e}+\Delta\mathbf{h}_{r,e}\right)\mathbf{r}^{(t)}\right)
\end{aligned} \tag{52}$$

Then, after performing an equivalence transformation, i.e., $\overline{\mathbf{G}}_e = \mathrm{diag}\left(\overline{\mathbf{h}}_{r,e}^{\dagger}\right)\mathbf{H}$, the first term on the right-hand side of (52) can be expanded to (53), as shown at the bottom of the page.

By the similar mathematical transformation, the second and third terms on the right-hand side of (37) can be equivalently expressed, respectively, as

$$\Im_2 = \mathbf{x}_e^H\mathbf{F}_i^{\dagger}\mathbf{x}_e + \widetilde{\mathbf{b}}_{2,i}^{\dagger}\mathbf{x}_e + \mathbf{x}_e^{\dagger}\widetilde{\mathbf{b}}_{1,i} + \widetilde{c}_i^*, \tag{54}$$

and

$$\Im_3 = \mathbf{x}_e^H\mathbf{Z}_i\mathbf{x}_e + \widetilde{\mathbf{d}}_i^{\dagger}\mathbf{x}_e + \mathbf{x}_e^{\dagger}\widetilde{\mathbf{d}}_i + \widetilde{e}_i. \tag{55}$$

Thus, the lower bound of (37) is $\Im_1 + \Im_2 + \Im_3$, and the proof is completed.

$$\begin{aligned}
\Im_1 = &\left(\overline{\mathbf{h}}_{s,e}^{\dagger} + \mathbf{r}^{(t),\dagger}\overline{\mathbf{G}}_e\right)\mathbf{w}_i^{(t)}\mathbf{w}_i^{\dagger}\left(\overline{\mathbf{h}}_{s,e} + \overline{\mathbf{G}}_e^{\dagger}\mathbf{r}\right) + \left(\overline{\mathbf{h}}_{s,e}^{\dagger} + \mathbf{r}^{(t),\dagger}\overline{\mathbf{G}}_e\right)\mathbf{w}_i^{(t)}\mathbf{w}_i^{\dagger}\Delta\mathbf{h}_{s,e} + vec^{\dagger}\left(\mathrm{diag}\left(\Delta\mathbf{h}_{r,e}^{\dagger}\right)\right) \\
&\times vec\left(\mathbf{r}\left(\overline{\mathbf{h}}_{s,e}^{\dagger} + \mathbf{r}^{(t),\dagger}\overline{\mathbf{G}}_e\right)\mathbf{w}_i^{(t)}\mathbf{w}_i^{\dagger}\mathbf{H}^{\dagger}\right) + \Delta\mathbf{h}_{s,e}^{\dagger}\mathbf{w}_i^{(t)}\mathbf{w}_i^{\dagger}\left(\overline{\mathbf{h}}_{s,e} + \overline{\mathbf{G}}_e^{\dagger}\mathbf{r}\right) + \Delta\mathbf{h}_{s,e}^{\dagger}\mathbf{w}_i^{(t)}\mathbf{w}_i^{\dagger}\Delta\mathbf{h}_{s,e} \\
&+ vec^{\dagger}\left(\mathrm{diag}\left(\Delta\mathbf{h}_{r,e}^{\dagger}\right)\right)\left(\mathbf{H}^*\mathbf{w}_i^*\mathbf{w}_i^{(t),T}\otimes\mathbf{r}\right)\Delta\mathbf{h}_{s,e}^* + vec^{\dagger}\left(\mathbf{r}^{(t)}\left(\overline{\mathbf{h}}_{s,e}^{\dagger} + \mathbf{r}^{\dagger}\overline{\mathbf{G}}_e\right)\mathbf{w}_i\mathbf{w}_i^{(t),\dagger}\mathbf{H}^{\dagger}\right) \\
&\times vec\left(\mathrm{diag}\left(\Delta\mathbf{h}_{r,e}^{\dagger}\right)\right) + \Delta\mathbf{h}_{s,e}^T\left(\mathbf{w}_i^*\mathbf{w}_i^{(t),T}\mathbf{H}^T\otimes\mathbf{r}^{(t),\dagger}\right)vec\left(\mathrm{diag}\left(\Delta\mathbf{h}_{r,e}^{\dagger}\right)\right) \\
&+ vec^{\dagger}\left(\mathrm{diag}\left(\Delta\mathbf{h}_{r,e}^{\dagger}\right)\right)\left(\mathbf{H}^*\mathbf{w}_i^*\mathbf{w}_i^{(t),T}\mathbf{H}^T\otimes\mathbf{r}\mathbf{r}^{(t),\dagger}\right)vec\left(\mathrm{diag}\left(\Delta\mathbf{h}_{r,e}^{\dagger}\right)\right) \\
= &\,\mathbf{x}_e^{\dagger}\mathbf{F}_i\mathbf{x}_e + \widetilde{\mathbf{b}}_{1,i}^{\dagger}\mathbf{x}_k + \mathbf{x}_e^{\dagger}\widetilde{\mathbf{b}}_{2,i} + \widetilde{c}_i
\end{aligned} \tag{53}$$

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Chengjun Jiang (Graduate Student Member, IEEE) received the B.S. degree in telecommunications engineering from the North University of China, Taiyuan, China, in 2019. He is currently pursuing the Ph.D. degree with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University. From November 2023 to November 2024, he was a Visiting Ph.D. Student with Nanyang Technological University, Singapore, under the supervision of Prof. Chau Yuen. His research interests include integrated sensing and communication, nonorthogonal multiple access, reconfigurable intelligent surface, physical layer security, and convex optimization.

Chensi Zhang (Member, IEEE) received the B.S. and Ph.D. degrees in telecommunications engineering from Xidian University, Xi'an, China, in 2010 and 2015, respectively. He is currently an Associate Professor with the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University. His research interests include deep learning, wireless sensing, cooperative communications, green communications, the key technology of 6G communication-sensing-computing integration, nonorthogonal multiple access, reconfigurable intelligent surface, integrated sensing and communication, and physical layer security.

Chongwen Huang (Member, IEEE) received the B.Sc. degree from Nankai University in 2010, the M.Sc. degree from the University of Electronic Science and Technology of China in 2013, and the Ph.D. degree from Singapore University of Technology and Design (SUTD) in 2019. From October 2019 to September 2020, he was a Post-Doctoral Researcher with SUTD. Since September 2020, he has been with Zhejiang University, as a tenure-track young Professor. His research interests include holographic MIMO surface/reconfigurable intelligent surface, B5G/6G wireless communications, mmWave/THz communications, and deep learning technologies for wireless communications. He was a recipient of the 2021 IEEE Marconi Prize Paper Award, the 2023 IEEE Fred W. Ellersick Prize Paper Award, and the 2021 IEEE ComSoc Asia–Pacific Outstanding Young Researcher Award. He has been serving as an Editor for IEEE COMMUNICATIONS LETTER, *Signal Processing* (Elsevier), *EURASIP Journal on Wireless Communications and Networking*, and *Physical Communication*, since 2021.

Jianhua Ge received the Ph.D. degree from the School of Telecommunications Engineering, Xidian University, Xi'an, China, in 1990. He is currently a Professor and the Deputy Director of the State Key Laboratory of Integrated Services Networks, School of Telecommunications Engineering, Xidian University. He has worked on digital television (DTV) standardization as a DTV technical expert. His research interests include digital communications, digital video broadcasting, and performance enhancement techniques for 4G/B4G cellular communication systems.

Dusit Niyato (Fellow, IEEE) received the B.Eng. degree from the King Mongkut's Institute of Technology Ladkrabang (KMITL), Thailand, and the Ph.D. degree in electrical and computer engineering from the University of Manitoba, Canada. He is currently a Professor with the College of Computing and Data Science, Nanyang Technological University, Singapore. His research interests include mobile generative AI, edge intelligence, decentralized machine learning, and incentive mechanism design.

Chau Yuen (Fellow, IEEE) received the B.Eng. and Ph.D. degrees from Nanyang Technological University, Singapore, in 2000 and 2004, respectively. He was a Post-Doctoral Fellow with Lucent Technologies Bell Labs, Murray Hill, in 2005. From 2006 to 2010, he was with the Institute for Infocomm Research, Singapore. From 2010 to 2023, he was with the Engineering Product Development Pillar, Singapore University of Technology and Design. Since 2023, he has been with the School of Electrical and Electronic Engineering, Nanyang Technological University. He is currently the Provost's Chair of Wireless Communications and the Assistant Dean of the Graduate College. He has four U.S. patents and published over 400 research papers at international journals. He received the IEEE Communications Society Leonard G. Abraham Prize (2024), the IEEE Communications Society Best Tutorial Paper Award (2024), the IEEE Communications Society Fred W. Ellersick Prize (2023), the IEEE Marconi Prize Paper Award in Wireless Communications (2021), the IEEE APB Outstanding Paper Award (2023), and the EURASIP Best Paper Award for *EURASIP Journal on Wireless Communications and Networking* (2021). He serves as the Editor-in-Chief for *Nature Computer Science* and an Editor for IEEE TRANSACTIONS ON VEHICULAR TECHNOLOGY, IEEE SYSTEMS JOURNAL, and IEEE TRANSACTIONS ON NETWORK SCIENCE AND ENGINEERING, where he was awarded as IEEE Transactions on Network Science and Engineering Excellent Editor Award and Top Associate Editor for TVT from 2009 to 2015. He served as the Guest Editor for several special issues, including IEEE JOURNAL ON SELECTED AREAS IN COMMUNICATIONS, *IEEE Wireless Communications Magazine*, *IEEE Communications Magazine*, *IEEE Vehicular Technology Magazine*, IEEE TRANSACTIONS ON COGNITIVE COMMUNICATIONS AND NETWORKING, and *Applied Energy* (Elsevier). He is a Distinguished Lecturer of IEEE Vehicular Technology Society, Top 2% Scientists by Stanford University, and also a Highly Cited Researcher by Clarivate Web of Science.

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